

8. Hydrogen peroxide solution, H_2O_2 , is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover A

Working temperature	= 50 °C
Cost per 100 cm ³	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm ³

Stain remover B

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm ³

Stain remover C

Working temperature	= 20 °C
Cost per 100 cm ³	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm ³

Stain remover D

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm ³

- (a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide

☐

type of oil used, towel material and temperature of stain remover

☐

type of oil used and towel material

☐

type of oil used, towel material and cost of stain remover

☐


- (b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover

☐

it is heat resistant

☐

it has a low concentration of hydrogen peroxide

☐

it takes a long time to work

☐

- (c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

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(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

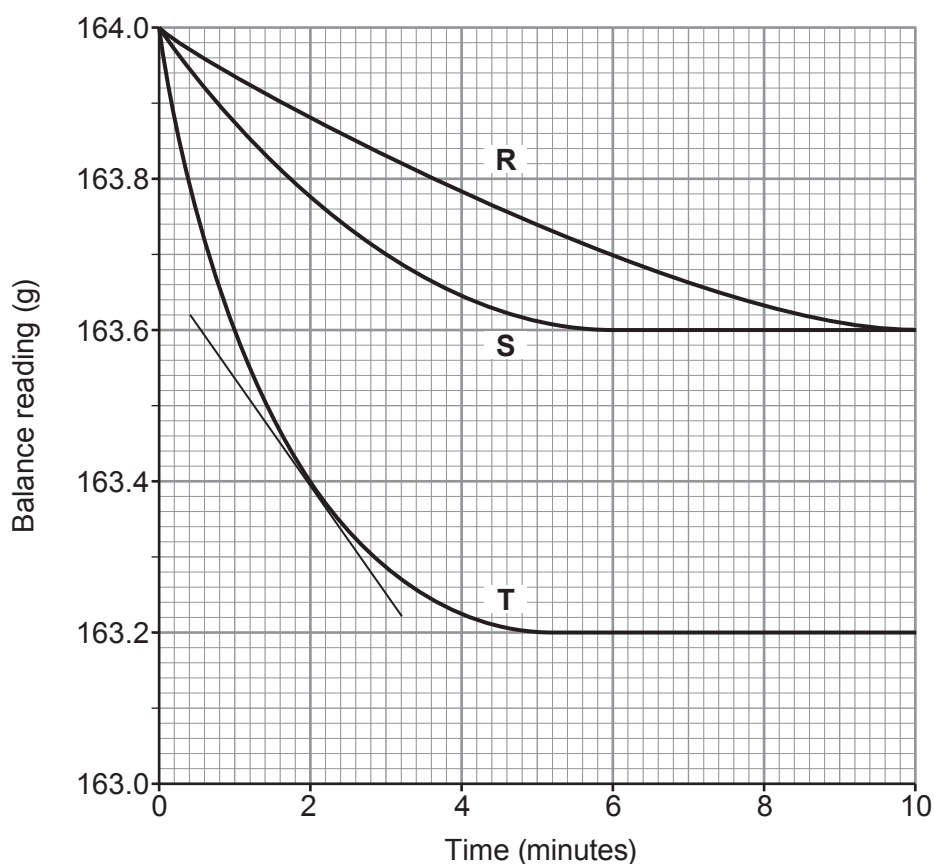


The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm³ of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



- (i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = g/minute

- (ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





20

THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1	H	
Hydrogen	1	

7	Li	9	Be
Lithium	3	Beryllium	4
23	Na	24	Mg
Sodium	11	Magnesium	12
39	K	40	Ca
Potassium	19	Calcium	20
86	Rb	88	Sr
Rubidium	37	Strontium	38
133	Cs	137	Ba
Caesium	55	Barium	56
223	Fr	226	Ra
Francium	87	Radium	88
		227	Ac
		Actinium	89

11	B	12	C	14	N	16	O	19	F	20	Ne
Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon	10
27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon	18
70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
Gallium	31	Germanium	32	Arsenic	33	Selenium	34	Bromine	35	Krypton	36
115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
Indium	49	Tin	50	Antimony	51	Tellurium	52	Iodine	53	Xenon	54
204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85	Radon	86

65	Zn	63.5	Cu	59	Ni	59	Co	56	Fe	55	Mn
Zinc	30	Copper	29	Nickel	28	Cobalt	27	Iron	26	Manganese	25
112	Cd	108	Ag	106	Pd	103	Rh	101	Ru	99	Tc
Cadmium	48	Silver	47	Palladium	46	Rhodium	45	Ruthenium	44	Technetium	43
201	Hg	197	Au	195	Pt	192	Ir	190	Os	186	Re
Mercury	80	Gold	79	Platinum	78	Iridium	77	Osmium	76	Rhenium	75
184	W	181	Ta	181	Ta	179	Hf	179	Hf	179	Hf
Tungsten	74	Tantalum	73	Tantalum	73	Hafnium	72	Hafnium	72	Hafnium	72
139	La	139	La	139	La	139	La	139	La	139	La
Lanthanum	57	Lanthanum	57	Lanthanum	57	Lanthanum	57	Lanthanum	57	Lanthanum	57

Key

Ar	Symbol	Name	Z

relative atomic mass

atomic number